

Convolutional Neural Network

Basic Concept

- Kernel = filter = feature detector
- Stride
- Padding
- Polling
- Flatten

Handwritten Digit Recognition





Location shifted

1	1	1	-1	-1
1	-1	1	-1	-1
1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
1	-1	-1	-1	-1



1	1	1	-1	-1
1	-1	1	-1	-1
1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
1	-1	-1	-1	-1



-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

Variation 1

-1	-1	1	-1	-1
-1	1	-1	1	-1
-1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1



-1	-1	1	-1	-1
-1	1	-1	1	-1
-1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1



-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
-1	1	-1	-1	-1

To handle **variety** in digits we can use simple artificial neural network (ANN)

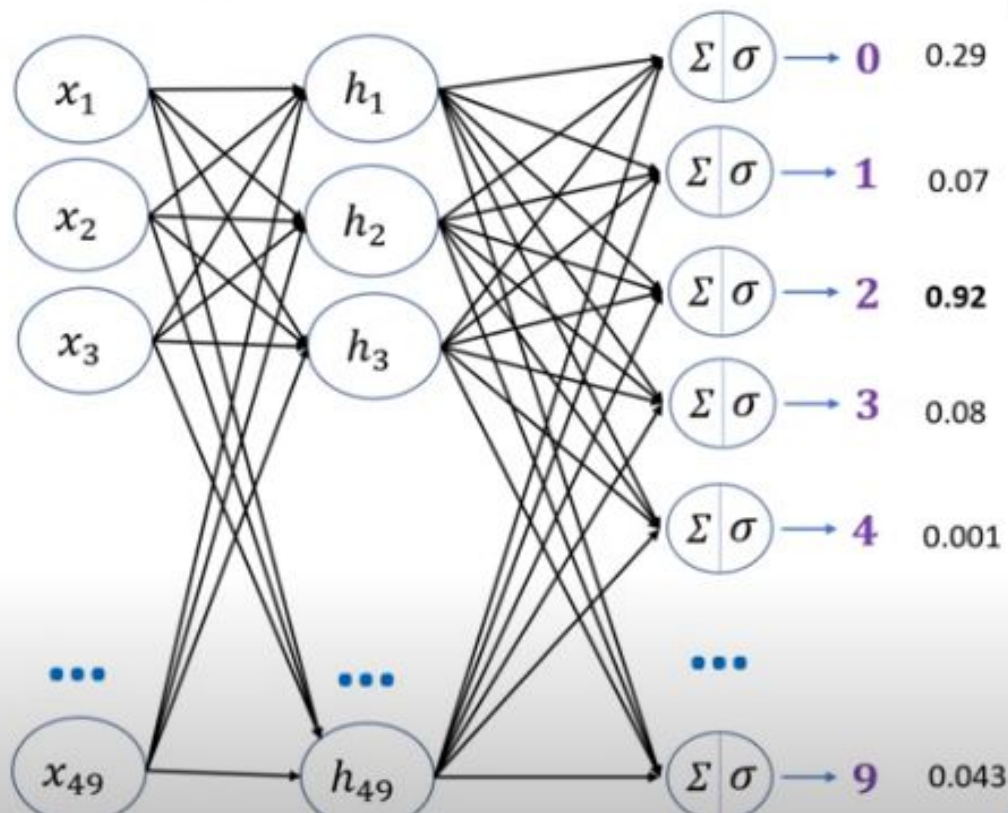




0	0	0	0	0	0	0
0	87	240	210	24	0	0
0	13	0	101	195	0	0
0	35	167	99	210	0	0
0	145	230	240	201	189	140
0	0	102	67	17	13	0
0	0	0	0	0	0	0

7 by 7 grid

0
0
0
0
0
0
0
0
0
87
240
210
24
0
...
0





channel

Image size = 1920 x 1080 X 3

Size of
image



Image size = $1920 \times 1080 \times 3$

First layer neurons = $1920 \times 1080 \times 3 \sim 6 \text{ million}$

Hidden layer neurons = Let's say you keep it $\sim 4 \text{ million}$

Weights between input and hidden layer = $6 \text{ mil} \times 4 \text{ mil}$
 $= 24 \text{ million}$

That is too much computations

Disadvantages of using ANN for image classification

1. Too much computation
2. Treats local pixels same as pixels far apart
3. Sensitive to location of an object in an image



Koala's **eye**? = Y



Koala's **nose**? = Y



Koala's **ears**? = Y



Koala's **head**? = Y



Koala's **hands**? = Y



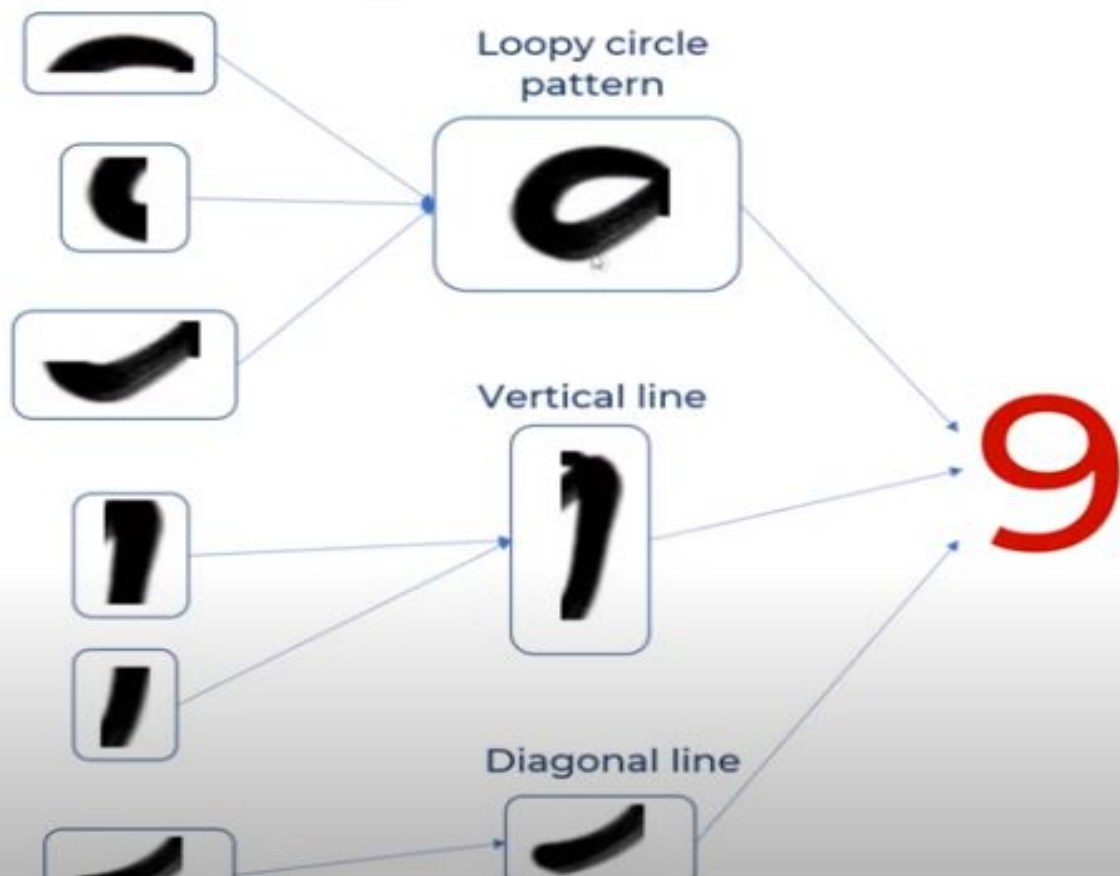
Koala's **legs**? = Y



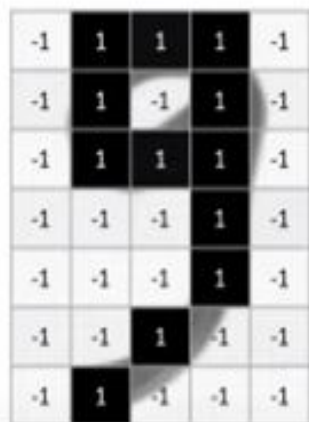
Koala's **body**? = Y



Is it **Koala**? = Y



Loopy pattern
filter



Vertical line
filter

Diagonal line
filter

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

*

1	1	1
1	-1	1
1	1	1

-0.11	1	-0.11
-0.55	0.11	-0.33
-0.33	0.33	-0.33
-0.22	-0.11	-0.22
-0.33	-0.33	-0.33

Feature Map

9

Loopy pattern detector

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ & & \\ & & \\ & & \\ & & \end{bmatrix}$$

6

Loopy pattern detector

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & & \\ & & \\ & 1 & \end{bmatrix}$$

8

Loopy pattern detector

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ & & \\ & & \\ & & \\ & 1 & \end{bmatrix}$$

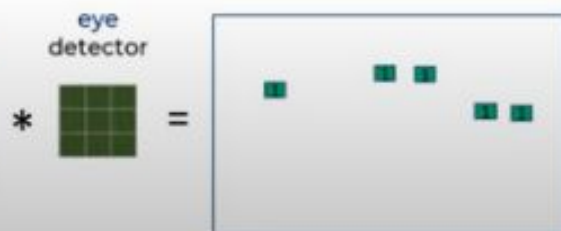
96

Loopy pattern detector

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ & & & \\ & & & \\ & & & \\ & & 1 & \end{bmatrix}$$

Filters are nothing
but the feature
detectors

Location invariant: It can detect eyes in any location of the image



Loopy pattern detector

$$\begin{array}{c} \text{9} \end{array} * \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ & & \\ & & \\ & & \\ & & \end{bmatrix}$$

Vertical line detector

$$\begin{array}{c} \text{9} \end{array} * \begin{bmatrix} -1 & 1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & 1 & \\ & & \\ & & \end{bmatrix}$$

Diagonal line detector

$$\begin{array}{c} \text{9} \end{array} * \begin{bmatrix} -1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & & \\ & 1 & \\ & & \end{bmatrix}$$

9

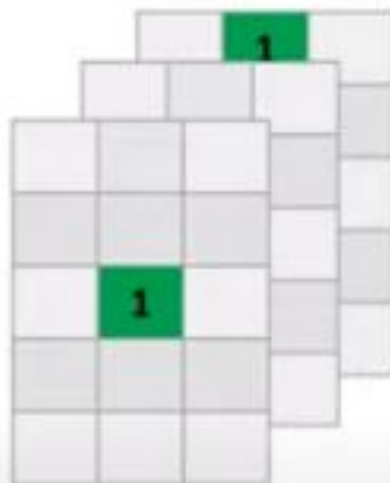
*

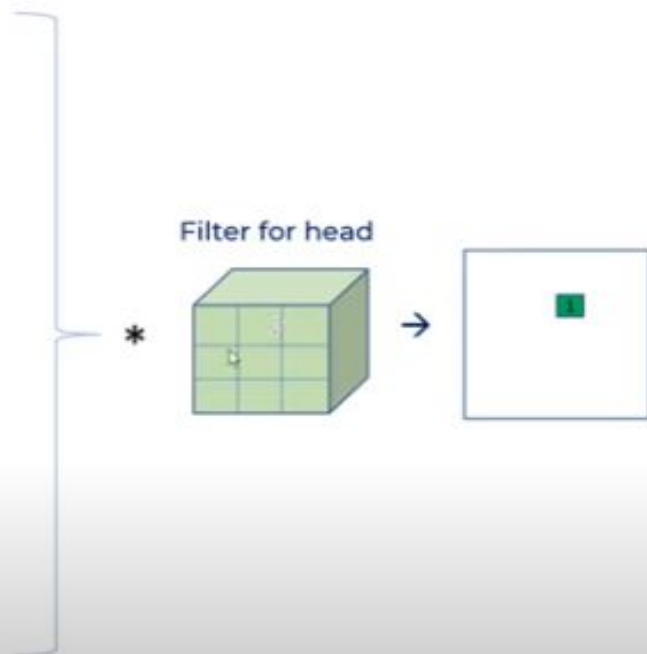
Filters

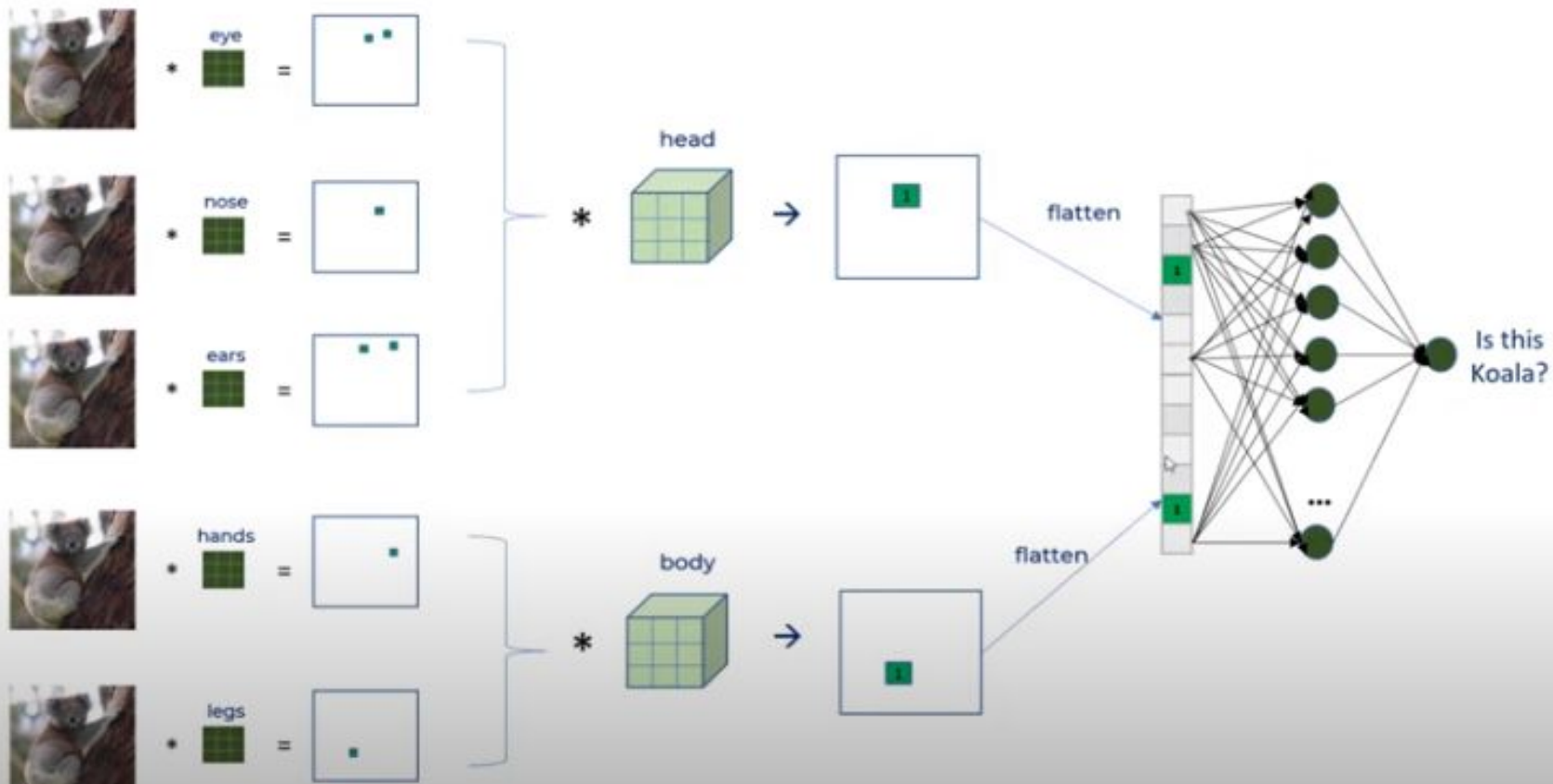


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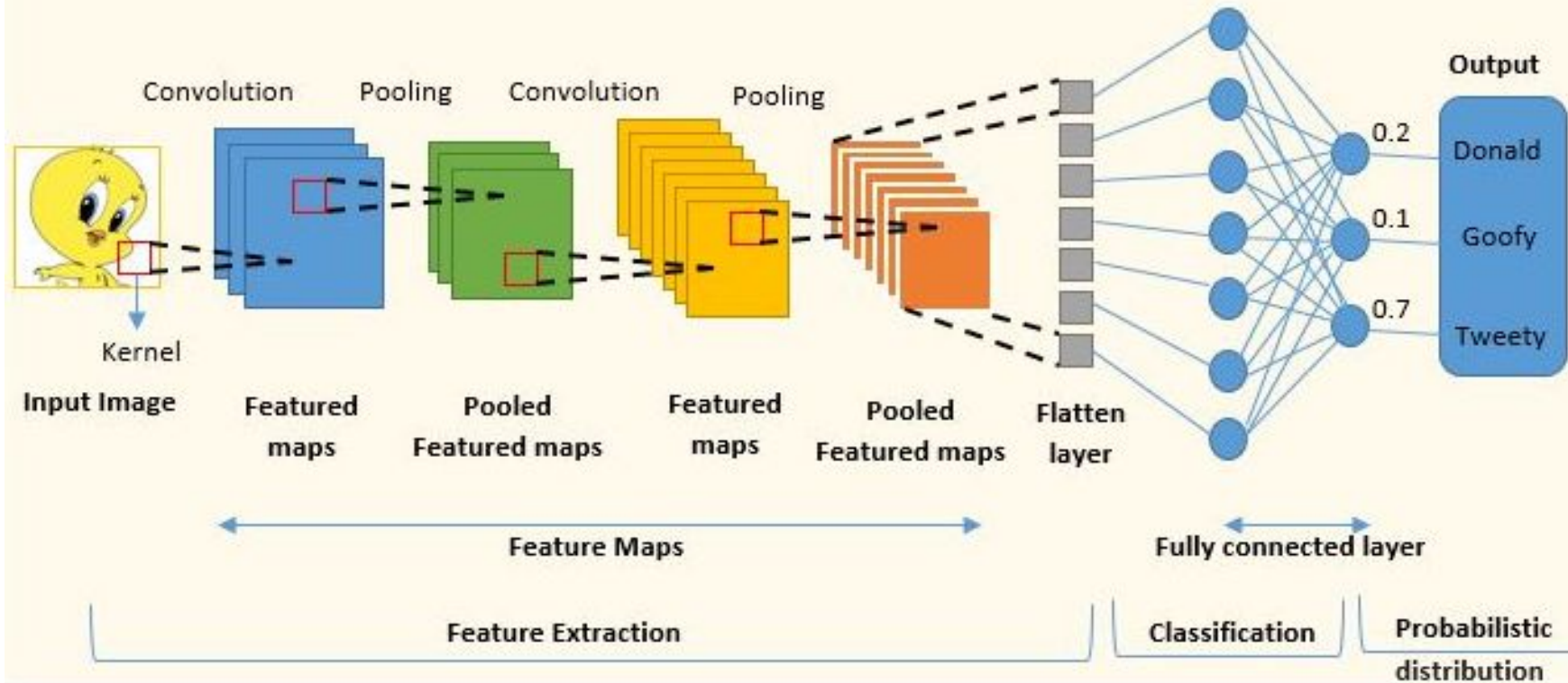
Feature Maps







A Typical Convolutional Neural Network (CNN)



Two Principles

1. Translation Invariance:

Our vision systems should, in some sense, **respond similarly to the same object regardless of where it appears** on the image.

Two Principles

2. Locality:

Our vision systems should, in some sense, **focus on somewhat local regions**, without regard for what else is happening on the image at greater distances.